

Application Program Development

Ch 03: Logical Operators and Boolean Variables

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Logical Operators

- **Logical operators** operators that can be used to create complex Boolean expressions
 - **and operator** or **operator**
 - connect two Boolean expressions into a compound Boolean expression
 - **not operator**
 - unary operator, reverses the truth of its Boolean operand

Logical Operators Concept

The logical and operator and the logical or operator allow you to connect multiple Boolean expressions to create a compound expression. The logical not operator reverses the truth of a Boolean expression.



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Logical Operator Truth Tables

A	B	A and B
False	False	False
False	True	False
True	False	False
True	True	True

A	B	A or B
False	False	False
False	True	True
True	False	True
True	True	True

A	not A
False	True
True	False



- **Short circuit evaluation:** deciding the value of a compound Boolean expression after evaluating only one sub expression
 - Performed by the or and and operators
 - Left to right order of evaluation
 - For or operator: as soon as operand from left to right evaluates as True you can stop, you know expression is True
 - For and operator: as soon as operand from left to right evaluates as False you can stop, you know expression is False



Compound Boolean Expressions using Logical Operators

Table 4: Compound Boolean expressions using logical operators

Expression	Meaning
$x > y$ and $a < b$	Is x greater than y AND is a less than b ?
$x == y$ or $x == z$	Is x equal to y OR is x equal to z ?
not ($x > y$)	Is the expression $x > y$ NOT true?
$x \geq 10$ and $x \leq 20$	Is x in the range 10 to 20 (inclusive)
$x < 10$ or $x > 20$	Is x not in the range 10 to 20 (inclusive)



Checking Numeric Ranges with Logical Operators

- Determine if a value is within a specific range
 - $x \geq 10$ and $x \leq 20$
- Determine whether a numeric value is outside of a specific range
 - $x < 10$ or $x > 20$

Note comparison operators have higher precedence, so don't need to put parentheses in these expressions (though will not hurt)

Note short-circuiting, for example if $x = 5$, and we are performing the first test to see if x is within the range, since $5 \geq 10$ is `False` the evaluation of this expression stops because we know the `and` will be `False` no matter what the right hand expression is.



Boolean Variables

Boolean Variables Concept

A Boolean variable can contain one of two values: True or False. Boolean variables are commonly used as flags, which indicate whether specific conditions exist or not.

- **Boolean variable:** contains one of two values, True or False
 - Represented by bool data type
- Commonly used as flags
 - **Flag:** variable that signals when some condition exists in a program
 - Flag set to False → condition does not exist
 - Flag set to True → condition exists

```
>>> sales = 25000
>>> sales_quota_met = sales >= 50000.0
>>> sales_quota_met
False
>>> type(sales_quota_met)
<class 'bool'>
```

```
>>> sales = 75000
>>> sales_quota_met = sales >= 50000.0
>>> sales_quota_met
True
>>> type(sales_quota_met)
<class 'bool'>
```



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Summary

- Logical operators `and` `or` and `not` allow you to connect multiple Boolean expressions to create compound expressions
 - This is useful to simplify and remove nesting of conditions
- A common use of Boolean Expressions is to check values are within or outside of needed range
- Python has a basic `bool` data type, it holds a single Boolean value, either `True` or `False`
 - The `bool` Boolean variable is useful as a **Flag** in a program



References



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